Technology Stack

Date	07 July 2026
Team ID	SWTID-2026-3218
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S. No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	HTML5, CSS3, Bootstrap, JavaScript	Provides a responsive and user-friendly web interface for entering applicant details and displaying prediction results.

2	**Backend / Server-Side**	Python 3.8+, Flask	Flask handles HTTP requests, processes user inputs, loads the trained machine learning model, and returns approval predictions efficiently.
3	**Database / Data Storage**	CSV Dataset (Pandas Data Frame)	Stores applicant and credit data for preprocessing, analysis, and model training. Pandas enables efficient data manipulation.
4	**Cloud / Hosting / Deployment**	IBM Watson Machine Learning (IBM Cloud)	Deploys the trained model on the cloud to provide scalable, real-time prediction services through the web application.
5	**Version Control & CI/CD**	Git, GitHub	Tracks source code changes, supports team collaboration, version management, and project sharing.
6	**Third-Party APIs / Other Tools**	Scikit-learn, XG Boost, NumPy, Pandas, Matplotlib, Seaborn, Anaconda Navigator,	Used for data preprocessing, visualization, machine learning model development, evaluation, and experimentation during the project lifecycle.

		Jupyter Notebook.	